

Solving Linear Equations with Variables on Both Sides

Answer Key

Grade 8 / Pre-Algebra

Quick Answers

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|------|-------------------|--------------|
| 1. 3 | 5. -3 | 9. $6x + 15$ |
| 2. 5 | 6. $\frac{11}{2}$ | 10. $4x + 8$ |
| 3. 4 | 7. $\frac{9}{4}$ | |
| 4. 8 | 8. 5 | |
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Curriculum

Level: Grade 8 / Pre-Algebra

8.EE.C.7 – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Difficulty 1–4 of 5 across 10 tagged checks.

1. Balance model: $3x + 2 = x + 8$.**Reading the picture.** The left pan holds 3 bags and 2 loose counters, so it is worth $3x + 2$. The right pan holds 1 bag and 8 loose counters, worth $x + 8$. The pans balance, so $3x + 2 = x + 8$.**Solving.** Take one bag off each pan — the same amount from both sides, so the balance holds:

$$3x + 2 - x = x + 8 - x \implies 2x + 2 = 8.$$

Now take 2 counters off each pan: $2x = 6$. Two equal bags are worth 6 counters, so one bag is worth 3.**Check.** $3(3) + 2 = 11$ and $3 + 8 = 11$ ✓. $\boxed{x = 3}$ **2.** $5x - 4 = 2x + 11$.**Solving.** Collect the variable terms on the left by subtracting $2x$ from both sides:

$$5x - 4 - 2x = 2x + 11 - 2x \implies 3x - 4 = 11.$$

Add 4 to both sides: $3x = 15$. Divide both sides by 3.**Check.** $5(5) - 4 = 21$ and $2(5) + 11 = 21$ ✓. $\boxed{x = 5}$

3. Balance model: $4x + 1 = 2x + 9$.

Reading the picture. The left pan holds 4 bags and 1 counter, worth $4x + 1$. The right pan holds 2 bags and 9 counters, worth $2x + 9$. Balanced pans give $4x + 1 = 2x + 9$.

Solving. Remove 2 bags from each pan:

$$4x + 1 - 2x = 2x + 9 - 2x \implies 2x + 1 = 9.$$

Remove 1 counter from each pan: $2x = 8$, so each bag holds 4 counters.

Check. $4(4) + 1 = 17$ and $2(4) + 9 = 17$ ✓. $\boxed{x = 4}$

4. $2(x + 5) = 4x - 6$.

Solving. The left side is a product, so distribute before anything else: $2(x + 5) = 2x + 10$. The equation becomes $2x + 10 = 4x - 6$. Subtract $2x$ from both sides:

$$10 = 2x - 6.$$

Add 6 to both sides: $16 = 2x$, so $x = 8$. Collecting on the right instead of the left kept the coefficient positive, which is why the arithmetic stayed simple.

Check. $2(8 + 5) = 26$ and $4(8) - 6 = 26$ ✓. $\boxed{x = 8}$

5. $8x + 1 = 3x - 14$.

Solving. Subtract $3x$ from both sides:

$$8x + 1 - 3x = 3x - 14 - 3x \implies 5x + 1 = -14.$$

Subtract 1 from both sides: $5x = -15$. Divide by 5.

Check. $8(-3) + 1 = -23$ and $3(-3) - 14 = -23$ ✓. $\boxed{x = -3}$

6. $3(2x - 1) = 4x + 8$.

Solving. Distribute on the left: $3(2x - 1) = 6x - 3$, so the equation is $6x - 3 = 4x + 8$. Subtract $4x$ from both sides:

$$2x - 3 = 8.$$

Add 3 to both sides: $2x = 11$. Divide by 2. The answer is a fraction, and that is fine — an equation does not have to have a whole-number solution.

Check. $3\left(2 \cdot \frac{11}{2} - 1\right) = 3(10) = 30$ and $4 \cdot \frac{11}{2} + 8 = 22 + 8 = 30$ ✓. $\boxed{x = \frac{11}{2}}$

7. $6x - 5 = 2x + 4$.

Solving. Subtract $2x$ from both sides:

$$6x - 5 - 2x = 2x + 4 - 2x \implies 4x - 5 = 4.$$

Add 5 to both sides: $4x = 9$. Divide by 4.

Check. $6 \cdot \frac{9}{4} - 5 = \frac{54}{4} - 5 = \frac{34}{4}$ and $2 \cdot \frac{9}{4} + 4 = \frac{18}{4} + 4 = \frac{34}{4}$ ✓. $\boxed{x = \frac{9}{4}}$

8. Two plans — the break-even point.

Setting it up. Plan A costs $40 + 12x$ dollars for x sessions and Plan B costs $20x$ dollars. “The two plans cost the same” is the sentence that makes an equation out of the table:

$$40 + 12x = 20x.$$

Solving. Subtract $12x$ from both sides: $40 = 8x$. Divide both sides by 8, giving $x = 5$ sessions.

Check. Plan A: $40 + 12(5) = 100$ dollars. Plan B: $20(5) = 100$ dollars ✓. Below 5 sessions Plan B is cheaper; above 5 sessions Plan A is. $x = 5$

9. $3(2x + 5) = 6x + 15$ — infinitely many solutions.

Solving. Distribute the left side: $3(2x + 5) = 6x + 15$. The equation is now $6x + 15 = 6x + 15$, the same expression on both sides. Subtract $6x$ from both sides and the variable disappears, leaving $15 = 15$ — a statement that is true no matter what x is.

Conclusion. Every real number satisfies the original equation; the two sides were never different expressions, only different-looking ones.

$$6x + 15 = 6x + 15 \implies \text{infinitely many solutions}$$

Grading note: accept “all real numbers”, “every number”, or “infinitely many solutions”. A student who writes $x = 0$ has stopped one step early.

10. Challenge: $4(x + 2) = 4x + 5$ — no solution.

Solving. Distribute the left side: $4(x + 2) = 4x + 8$. The equation is now $4x + 8 = 4x + 5$. Subtract $4x$ from both sides and the variable disappears, leaving $8 = 5$ — a statement that is false no matter what x is.

Conclusion. No number makes the original equation true, so the solution set is empty.

$$4x + 8 = 4x + 5 \implies \text{no solution}$$

Grading the justification: accept any explanation of the form “both sides grow at the same rate, so the gap between them never closes” or “the variable cancelled and left a false number sentence”. Contrast it with problem 9: there the leftover sentence was true, so everything worked; here it is false, so nothing does.